EGFR fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes EGFR gene fusions from the msk_impact_50k_2026 study, covering 55 fusion events. 28/40 fusions with known frame status (70.0%) were in-frame. 41/55 fusions (74.5%) retained the Protein tyrosine and serine/threonine kinase domain. Domain retention was tested with Fisher's exact test (p=0.018443, statistically significant at alpha=0.05). No literature benchmark is configured for EGFR.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 55 analyzed fusion events, identifying 31 distinct fusion partner genes. The most frequent partner was SEPTIN14, observed in 18 events.

Domain retention

Protein tyrosine and serine/threonine kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 25/28 (89.3%) of in-frame fusions retained the domain, p=0.018443 (statistically significant at alpha=0.05). The Protein tyrosine and serine/threonine kinase appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 9.9999e-06.

Domain disruption

Disruption of the Receptor L domain, Furin-like cysteine rich region, and Growth factor receptor domain IV was tested with Fisher's exact test comparing in-frame fusions against all others; 6/27 (22.2%) of in-frame fusions disrupted the domain, p=0.990471 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

Junction concentration

48% of 31 fusions with this gene as the 5' partner join at exon 24, against 0% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=9.9999e-06, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 49 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 981 aa (exon 24; label-permutation p=0.0889111). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is 16 aa from the nearest configured domain boundary at 965 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=37) and "not_retained" (n=12). For Frame_status, retained: 24/30 (80.0%, 95% CI 62.7%-90.5%); not_retained: 3/8 (37.5%, 95% CI 13.7%-69.4%). Welch's t-test comparing Tumor_variant_count between groups gave means of 388.892 vs 208.833, p=0.34888 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: cutpoint_proximity sub-score excluded: corrected cutpoint p-value was missing or >= 0.05, cutpoint was not determinable, or mapped breakpoints were unavailable.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for EGFR in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (14 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 6 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: EGFR has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run (3 row(s) in retention_counter_intuitive_events); see the accompanying table.

Mutation cooccurrence

Mutation cooccurrence reported: EGFR has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (111 row(s) in window_scan, 58 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SEPTIN14	18	0.32727272727272727	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.3310717248920916	1
LANCL2	3	0.05454545454545454	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.23481504040011297	2
NIPSNAP2	3	0.05454545454545454	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.23481504040011297	3
FADD	2	0.03636363636363636	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.2283979281006477	4
RAD51	2	0.03636363636363636	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.2283979281006477	5
VWC2	2	0.03636363636363636	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.2283979281006477	6
BLTP3B	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	7
CADPS	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	8
CDH7	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	9
CSF2RA	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	10
DDC	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EEA1	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	12
ELAPOR2	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	13
GARS1	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	14
KIF5B	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	15
NUMA1	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	16
PCDH15	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	17
PDE1C	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	18
PKD1L1	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	19
PLOD3	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	20
SCAF4	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	21
SEL1L	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	22
TNRC18	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	23
TNS3	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TUT7	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	25
VOPP1	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	26
VPS41	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	27
VSTM2A	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	28
YIF1B	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	29
ZNF713	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	30
ZPBP	1	0.01818181818181818	0.5000004342923104	0.0	None	0.5822903940338204	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22198081580118248	31

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
30	1	36	0	12	None	1.0	-0.0
142	2	35	0	12	None	1.0	-0.0
249	2	35	1	11	0.6285714285714286	1.0	-0.0
288	2	35	2	10	0.2857142857142857	0.2482159376238932	0.605170336455772
297	6	31	2	10	0.967741935483871	1.0	-0.0
336	7	30	2	10	1.1666666666666667	1.0	-0.0
627	7	30	3	9	0.7	0.6899264605876614	0.16119719833962448
640	8	29	3	9	0.8275862068965517	1.0	-0.0
651	9	28	3	9	0.9642857142857143	1.0	-0.0
687	9	28	5	7	0.45	0.2850257317974849	0.5451159306118377
688	13	24	5	7	0.7583333333333333	0.7383641737145339	0.1317293839425359

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
823	13	24	6	6	0.5416666666666666	0.497546681993718	0.3031661656093887
875	13	24	7	5	0.3869047619047619	0.18903503442605862	0.7234576992771142
876	13	24	8	4	0.2708333333333333	0.092236691980081	1.0350962811747513
981	13	24	9	3	0.18055555555555555	0.021740254820221648	1.6627353698004341
982	28	9	9	3	1.037037037037037	1.0	-0.0
983	28	9	11	1	0.2828282828282828	0.4143055366655828	0.38267926287056014
986	29	8	11	1	0.32954545454545453	0.42030696000904744	0.37643341830173344
994	30	7	11	1	0.38961038961038963	0.6597711605779072	0.18060667193042876
1003	31	6	11	1	0.4696969696969697	0.6650872361938774	0.17712138670765892
1038	32	5	11	1	0.5818181818181818	1.0	-0.0
1054	35	2	11	1	1.5909090909090908	1.0	-0.0
1055	35	2	12	0	0.0	1.0	-0.0
1075	36	1	12	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

6	11
21	11

frame_domain_contingency_tables_by_tumor_type

stratum	table
ARMM	[0, 1], [0, 0]
BLCA	[0, 0], [0, 1]
BRCA	[0, 0], [0, 1]
CCRCC	[0, 0], [0, 1]
COAD	[2, 0], [0, 0]
DDLS	[0, 1], [0, 0]
ESCA	[0, 0], [1, 0]
GB	[2, 4], [16, 3]
IAMPCA	[0, 0], [1, 0]
IDC	[0, 2], [1, 0]
LNET	[0, 0], [0, 1]
LUAD	[1, 0], [1, 1]

stratum	table
LUSC	[0, 1], [0, 1]
MPNST	[0, 0], [0, 1]
NSGCT	[0, 1], [0, 0]
OS	[0, 1], [0, 0]
PAAC	[1, 0], [0, 0]
STAD	[0, 0], [1, 0]
UMEC	[0, 0], [0, 1]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf01030	Receptor L domain	49	32	1	16	0.01377848436671966
auto_disruption_pf00757	Furin-like cysteine rich region	49	31	7	11	0.12265512265512266
auto_disruption_pf14843	Growth factor receptor domain IV	49	35	1	13	0.06655844155844155

domain_retention tables

frame_domain_contingency_table

24	13
3	9

frame_domain_contingency_tables_by_tumor_type

stratum	table
ARMM	[0, 1], [0, 0]
BLCA	[0, 0], [0, 1]
BRCA	[0, 1], [0, 0]
CCRCC	[0, 0], [0, 1]
COAD	[2, 0], [0, 0]
DDLS	[0, 1], [0, 0]
ESCA	[1, 0], [0, 0]
GB	[16, 4], [2, 3]
IAMPCA	[1, 0], [0, 0]
IDC	[0, 1], [1, 1]
LNET	[0, 0], [0, 1]
LUAD	[2, 1], [0, 0]

stratum	table
LUSC	[0, 0], [0, 2]
MPNST	[0, 1], [0, 0]
NSGCT	[0, 1], [0, 0]
OS	[0, 1], [0, 0]
PAAC	[1, 0], [0, 0]
STAD	[1, 0], [0, 0]
UMEC	[0, 1], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	49	37	3	9	0.11989459815546773

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
18	39	49	0.7959183673469388

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
BLTP3B	1	1	1	1
CADPS	1	1	1	1
CDH7	1	1	1	1
CSF2RA	1	1	1	1
DDC	1	1	1	1
EEA1	1	1	1	1
ELAPOR2	1	1	1	1
FADD	2	2	2	1
GARS1	1	1	1	1
KIF5B	1	1	1	1
LANCL2	3	3	3	3

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
NIPSNAP2	3	3	3	2
NUMA1	1	1	1	1
PCDH15	1	1	1	1
PDE1C	1	1	1	1
PKD1L1	1	1	1	1
PLOD3	1	1	1	1
RAD51	2	2	2	1
SCAF4	1	1	1	1
SEL1L	1	1	1	1
SEPTIN14	18	18	18	18
TNRC18	1	1	1	1
TNS3	1	1	1	1
TUT7	1	1	1	1
VOPP1	1	1	1	1
VPS41	1	1	1	1
VSTM2A	1	1	1	1
VWC2	2	2	2	2
YIF1B	1	1	1	1
ZNF713	1	1	1	1
ZPBP	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
7	55268000	55269000	21	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (21 total)
7	55241000	55242000	6	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (6 total)
7	55269000	55270000	5	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526

chromosome	bin_start	bin_end	n_events	event_ids
7	55222000	55223000	4	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
7	55238000	55239000	2	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558
7	55259000	55260000	2	EVT-P-0030222-T01-IM6-557, EVT-P-0059222-T03-IM7-501
7	55270000	55271000	2	EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-516
7	55179000	55180000	1	EVT-P-0059032-T01-IM7-546
7	55214000	55215000	1	EVT-P-0028484-T02-IM6-548
7	55220000	55221000	1	EVT-P-0016009-T01-IM6-528
7	55221000	55222000	1	EVT-P-0063142-T01-IM7-503
7	55224000	55225000	1	EVT-P-0038683-T01-IM6-4
7	55240000	55241000	1	EVT-P-0065470-T01-IM7-545
7	55254000	55255000	1	EVT-P-0064660-T01-IM7-9

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
982	15	15	700	False
297	4	4	281	False
688	4	4	476	False
1054	3	3	407	False
687	2	2	432	False
983	2	2	93	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	24	15	31	0.4838709677419355	0.004529211335494058	26.71132292027409
five_prime	25	4	31	0.12903225806451613	0.0028411451840771214	5.714847897469783
three_prime	8	4	18	0.2222222222222222	0.009318125155821491	4.682364988143231
three_prime	18	4	18	0.2222222222222222	0.004773331671237431	5.822230246965958
five_prime	26	3	31	0.0967741935483871	0.004061746862793984	3.558129376510072
five_prime	17	2	31	0.06451612903225806	0.004872018615474113	1.9978675807412996

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	27	2	31	0.06451612903225806	0.014226502119172276	1.1443176416173815
three_prime	17	2	18	0.1111111111111111	0.009925828970331589	1.867618129117636
three_prime	25	2	18	0.1111111111111111	0.004892794814260783	2.4588273792190134
five_prime	6	1	31	0.03225806451612903	0.007609282805617884	0.6760411018090037
five_prime	7	1	31	0.03225806451612903	0.009447976398238179	0.5935758350345745
five_prime	15	1	31	0.03225806451612903	0.030618922961854898	0.20855924820421692
five_prime	20	1	31	0.03225806451612903	0.05415316213745533	0.08513366160555043
five_prime	21	1	31	0.03225806451612903	0.00543817003241087	0.8081922783087642
three_prime	2	1	18	0.05555555555555555	0.6392420842682622	4.652324068037482e-09
three_prime	4	1	18	0.05555555555555555	0.016891049613562702	0.5782619647695091
three_prime	9	1	18	0.05555555555555555	0.0037033574337239257	1.1897477401577186
three_prime	22	1	18	0.05555555555555555	0.00502264605667747	1.0622345704043525
three_prime	24	1	18	0.05555555555555555	0.008050777029834621	0.8683476018235466
three_prime	27	1	18	0.05555555555555555	0.004378583894290701	1.1194803212281925

mechanistic_interpretation tables

retention_counter_intuitive_events

event_id	sample_id	partner_gene	breakpoint_protein_position	domain_status
EVT-P-0059222-T03-IM7-501	P-0059222-T03-IM7	LANCL2	875	disrupted
EVT-P-0029159-T01-IM6-508	P-0029159-T01-IM6	SEPTIN14	1055	lost
EVT-P-0016009-T01-IM6-528	P-0016009-T01-IM6	VOPP1	249	lost

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
97	297	200	7	42	5	32	0.78125	1.0	-0.0	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0028484-T02-IM6-548, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, ... (7 total)
136	336	200	8	41	6	31	0.967741935483871	1.0	-0.0	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0028484-T02-IM6-548, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
142	342	200	8	41	6	31	0.967741935483871	1.0	-0.0	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0028484-T02-IM6-548, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, ... (8 total)
197	297	100	6	43	4	33	0.6060606060606061	0.6259415884762788	0.2034661923263583	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
236	336	100	7	42	5	32	0.78125	1.0	-0.0	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, ... (7 total)
247	297	50	6	43	4	33	0.6060606060606061	0.6259415884762788	0.2034661923263583	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
249	299	50	6	43	4	33	0.6060606060606061	0.6259415884762788	0.2034661923263583	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
249	349	100	7	42	5	32	0.78125	1.0	-0.0	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, ... (7 total)
249	449	200	7	42	5	32	0.78125	1.0	-0.0	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, ... (7 total)
272	297	25	5	44	4	33	1.3333333333333333	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, EVT-P-0063743-T01-IM7-536
286	336	50	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
288	313	25	5	44	4	33	1.3333333333333333	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, EVT-P-0063743-T01-IM7-536
288	338	50	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
288	388	100	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
288	488	200	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
297	322	25	4	45	4	33	None	0.5599312805603278	0.2518652699665775	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
297	347	50	5	44	5	32	None	0.31544865864939875	0.5010713149608126	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
297	397	100	5	44	5	32	None	0.31544865864939875	0.5010713149608126	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
297	497	200	5	44	5	32	None	0.31544865864939875	0.5010713149608126	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
487	687	200	5	44	2	35	0.17142857142857143	0.08685740716268008	1.0611931391891667	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558, EVT-P-0027944-T01-IM6-535, EVT-P-0039262-T01-IM6-529, EVT-P-0065470-T01-IM7-545
488	688	200	9	40	6	31	0.5806451612903226	0.6692548912935057	0.1744084460595711	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, ... (9 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
587	687	100	5	44	2	35	0.17142857142857143	0.08685740716268008	1.0611931391891667	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558, EVT-P-0027944-T01-IM6-535, EVT-P-0039262-T01-IM6-529, EVT-P-0065470-T01-IM7-545
588	688	100	9	40	6	31	0.5806451612903226	0.6692548912935057	0.1744084460595711	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, ... (9 total)
623	823	200	10	39	6	31	0.3870967741935484	0.2326462913533036	0.633303866223915	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, ... (10 total)
627	727	100	9	40	6	31	0.5806451612903226	0.6692548912935057	0.1744084460595711	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, ... (9 total)
627	827	200	10	39	6	31	0.3870967741935484	0.2326462913533036	0.633303866223915	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, ... (10 total)
637	687	50	4	45	2	35	0.2857142857142857	0.2482159376238932	0.605170336455772	EVT-P-0019922-T01-IM6-558, EVT-P-0027944-T01-IM6-535, EVT-P-0039262-T01-IM6-529, EVT-P-0065470-T01-IM7-545
638	688	50	8	41	6	31	0.967741935483871	1.0	-0.0	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, ... (8 total)
640	690	50	8	41	6	31	0.967741935483871	1.0	-0.0	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, ... (8 total)
640	740	100	8	41	6	31	0.967741935483871	1.0	-0.0	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
640	840	200	9	40	6	31	0.5806451612903226	0.6692548912935057	0.1744084460595711	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, ... (9 total)
651	701	50	7	42	5	32	0.78125	1.0	-0.0	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (7 total)
651	751	100	7	42	5	32	0.78125	1.0	-0.0	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (7 total)
651	851	200	8	41	5	32	0.46875	0.3858206664977805	0.41361451306963637	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (8 total)
663	688	25	6	43	4	33	0.6060606060606061	0.6259415884762788	0.2034661923263583	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (6 total)
675	875	200	8	41	4	33	0.24242424242424243	0.08756811245893278	1.057654011460585	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (8 total)
676	876	200	9	40	4	33	0.1696969696969697	0.02922094106413788	1.5343058016672584	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0030222-T01-IM6-557, EVT-P-0036028-T01-IM6-543, ... (9 total)
687	712	25	6	43	4	33	0.6060606060606061	0.6259415884762788	0.2034661923263583	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (6 total)
687	737	50	6	43	4	33	0.6060606060606061	0.6259415884762788	0.2034661923263583	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
687	787	100	6	43	4	33	0.6060606060606061	0.6259415884762788	0.2034661923263583	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (6 total)
687	887	200	9	40	4	33	0.1696969696969697	0.02922094106413788	1.5343058016672584	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0030222-T01-IM6-557, EVT-P-0036028-T01-IM6-543, ... (9 total)
688	713	25	4	45	4	33	None	0.5599312805603278	0.2518652699665775	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3
688	738	50	4	45	4	33	None	0.5599312805603278	0.2518652699665775	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3
688	788	100	4	45	4	33	None	0.5599312805603278	0.2518652699665775	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3
688	888	200	7	42	4	33	0.36363636363636365	0.3403233440182433	0.46810826004144224	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0030222-T01-IM6-557, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3, ... (7 total)
781	981	200	4	45	0	37	0.0	0.002336272159187449	2.6314765663762176	EVT-P-0030222-T01-IM6-557, EVT-P-0034892-T01-IM6-497, EVT-P-0059222-T03-IM7-501, EVT-P-0064660-T01-IM7-9
782	982	200	19	30	15	22	1.3636363636363635	0.7434280929329625	0.12876103165131966	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0028241-T01-IM6-518, ... (19 total)
783	983	200	21	28	15	22	0.6818181818181818	0.7388569842438371	0.1314396170375779	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0025018-T01-IM6-539, ... (21 total)
786	986	200	22	27	16	21	0.7619047619047619	0.7465376611778418	0.12694827818791946	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (22 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
794	994	200	23	26	17	20	0.85	1.0	-0.0	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (23 total)
803	1003	200	24	25	18	19	0.9473684210526315	1.0	-0.0	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (24 total)
823	1023	200	24	25	18	19	0.9473684210526315	1.0	-0.0	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (24 total)
838	1038	200	24	25	19	18	1.4777777777777779	0.7416433266833333	0.12980490670896044	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (24 total)
854	1054	200	27	22	22	15	2.0533333333333332	0.3311410503304157	0.4799869780310702	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (27 total)
855	1055	200	28	21	22	15	1.4666666666666666	0.7388569842438368	0.13143961703757803	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (28 total)
875	1075	200	29	20	23	14	1.6428571428571428	0.5123352016371653	0.2904458034998809	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (29 total)
876	1076	200	28	21	23	14	2.3	0.3156633225694666	0.5007758765526734	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (28 total)
882	982	100	16	33	15	22	7.5	0.0732677895601768	1.1350869106351265	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0028241-T01-IM6-518, ... (16 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
883	983	100	18	31	15	22	2.0454545454545454	0.4941707358480989	0.3061229765300417	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0025018-T01-IM6-539, ... (18 total)
886	986	100	19	30	16	21	2.2857142857142856	0.3229352452397617	0.49088455337919445	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (19 total)
891	1091	200	28	21	24	13	3.6923076923076925	0.09223669198008103	1.035096281174751	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (28 total)
894	994	100	20	29	17	20	2.55	0.31277927391835814	0.5047620328535949	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (20 total)
903	1003	100	21	28	18	19	2.8421052631578947	0.1918010226060691	0.7171490816753184	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (21 total)
932	982	50	16	33	15	22	7.5	0.0732677895601768	1.1350869106351265	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0028241-T01-IM6-518, ... (16 total)
933	983	50	18	31	15	22	2.0454545454545454	0.4941707358480989	0.3061229765300417	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0025018-T01-IM6-539, ... (18 total)
936	986	50	19	30	16	21	2.2857142857142856	0.3229352452397617	0.49088455337919445	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (19 total)
938	1038	100	22	27	19	18	3.1666666666666665	0.1819164698983225	0.7401279800890251	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (22 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
944	994	50	20	29	17	20	2.55	0.31277927391835814	0.5047620328535949	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (20 total)
953	1003	50	21	28	18	19	2.8421052631578947	0.1918010226060691	0.7171490816753184	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (21 total)
954	1054	100	25	24	22	15	4.4	0.050671040190493594	1.2952401802888702	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (25 total)
955	1055	100	26	23	22	15	2.933333333333333	0.18347427915301484	0.7364248099124756	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (26 total)
957	982	25	16	33	15	22	7.5	0.0732677895601768	1.1350869106351265	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0028241-T01-IM6-518, ... (16 total)
958	983	25	18	31	15	22	2.0454545454545454	0.4941707358480989	0.3061229765300417	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0025018-T01-IM6-539, ... (18 total)
961	986	25	19	30	16	21	2.2857142857142856	0.3229352452397617	0.49088455337919445	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (19 total)
969	994	25	20	29	17	20	2.55	0.31277927391835814	0.5047620328535949	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (20 total)
975	1075	100	27	22	23	14	3.2857142857142856	0.10368735297790324	0.9842742124331854	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (27 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
978	1003	25	21	28	18	19	2.8421052631578947	0.1918010226060691	0.7171490816753184	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (21 total)
981	1006	25	21	28	18	19	2.8421052631578947	0.1918010226060691	0.7171490816753184	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (21 total)
981	1031	50	21	28	18	19	2.8421052631578947	0.1918010226060691	0.7171490816753184	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (21 total)
981	1081	100	27	22	23	14	3.2857142857142856	0.10368735297790324	0.9842742124331854	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (27 total)
981	1181	200	28	21	24	13	3.6923076923076925	0.09223669198008103	1.035096281174751	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (28 total)
982	1007	25	20	29	18	19	4.7368421052631575	0.08925707056665505	1.0493573708857438	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (20 total)
982	1032	50	20	29	18	19	4.7368421052631575	0.08925707056665505	1.0493573708857438	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (20 total)
982	1082	100	26	23	23	14	4.928571428571429	0.04405035085805119	1.356050628112121	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (26 total)
982	1182	200	27	22	24	13	5.538461538461538	0.02174025482022164	1.6627353698004343	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (27 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
983	1008	25	5	44	3	34	0.4411764705882353	0.5843795427514207	0.23330499572487204	EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0037526-T01-IM6-525, EVT-P-0037759-T01-IM6-542
983	1033	50	5	44	3	34	0.4411764705882353	0.5843795427514207	0.23330499572487204	EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0037526-T01-IM6-525, EVT-P-0037759-T01-IM6-542
983	1083	100	11	38	8	29	0.8275862068965517	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0025191-T02-IM6-522, ... (11 total)
983	1183	200	12	37	9	28	0.9642857142857143	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0025191-T02-IM6-522, ... (12 total)
986	1086	100	9	40	8	29	3.0344827586206895	0.42030696000904744	0.37643341830173344	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, ... (9 total)
986	1186	200	10	39	9	28	3.5357142857142856	0.4143055366655828	0.38267926287056014	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, ... (10 total)
991	1091	100	9	40	8	29	3.0344827586206895	0.42030696000904744	0.37643341830173344	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, ... (9 total)
994	1094	100	9	40	8	29	3.0344827586206895	0.42030696000904744	0.37643341830173344	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, ... (9 total)
994	1194	200	9	40	8	29	3.0344827586206895	0.42030696000904744	0.37643341830173344	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, ... (9 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1003	1103	100	8	41	7	30	2.566666666666667	0.6597711605779072	0.18060667193042876	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, ... (8 total)
1003	1203	200	8	41	7	30	2.566666666666667	0.6597711605779072	0.18060667193042876	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, ... (8 total)
1004	1054	50	4	45	4	33	None	0.5599312805603278	0.2518652699665775	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
1005	1055	50	5	44	4	33	1.3333333333333333	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
1025	1075	50	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0043097-T01-IM6-531, ... (6 total)
1029	1054	25	4	45	4	33	None	0.5599312805603278	0.2518652699665775	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
1030	1055	25	5	44	4	33	1.3333333333333333	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
1038	1063	25	5	44	4	33	1.3333333333333333	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
1038	1088	50	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0043097-T01-IM6-531, ... (6 total)
1038	1138	100	7	42	6	31	2.129032258064516	0.6650872361938774	0.17712138670765892	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-516, ... (7 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1038	1238	200	7	42	6	31	2.129032258064516	0.6650872361938774	0.17712138670765892	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-516, ... (7 total)
1041	1091	50	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-516, ... (6 total)
1050	1075	25	5	44	4	33	1.3333333333333333	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0058375-T01-IM6-526
1054	1079	25	5	44	4	33	1.3333333333333333	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0058375-T01-IM6-526
1054	1104	50	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-516, ... (6 total)
1054	1154	100	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-516, ... (6 total)
1054	1254	200	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-516, ... (6 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
781	981	200	4	45	0	37	0.0	0.002336272159187449	2.6314765663762176	EVT-P-0030222-T01-IM6-557, EVT-P-0034892-T01-IM6-497, EVT-P-0059222-T03-IM7-501, EVT-P-0064660-T01-IM7-9
982	1182	200	27	22	24	13	5.538461538461538	0.02174025482022164	1.6627353698004343	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (27 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
676	876	200	9	40	4	33	0.1696969696969697	0.02922094106413788	1.5343058016672584	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0030222-T01-IM6-557, EVT-P-0036028-T01-IM6-543, ... (9 total)
982	1082	100	26	23	23	14	4.928571428571429	0.04405035085805119	1.356050628112121	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (26 total)
954	1054	100	25	24	22	15	4.4	0.050671040190493594	1.2952401802888702	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (25 total)
957	982	25	16	33	15	22	7.5	0.0732677895601768	1.1350869106351265	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0028241-T01-IM6-518, ... (16 total)
587	687	100	5	44	2	35	0.17142857142857143	0.08685740716268008	1.0611931391891667	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558, EVT-P-0027944-T01-IM6-535, EVT-P-0039262-T01-IM6-529, EVT-P-0065470-T01-IM7-545
675	875	200	8	41	4	33	0.24242424242424243	0.08756811245893278	1.057654011460585	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (8 total)
982	1007	25	20	29	18	19	4.7368421052631575	0.08925707056665505	1.0493573708857438	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (20 total)
891	1091	200	28	21	24	13	3.6923076923076925	0.09223669198008103	1.035096281174751	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (28 total)
975	1075	100	27	22	23	14	3.2857142857142856	0.10368735297790324	0.9842742124331854	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (27 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
938	1038	100	22	27	19	18	3.1666666666666665	0.1819164698983225	0.7401279800890251	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (22 total)
955	1055	100	26	23	22	15	2.933333333333333	0.18347427915301484	0.7364248099124756	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (26 total)
978	1003	25	21	28	18	19	2.8421052631578947	0.1918010226060691	0.7171490816753184	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (21 total)
623	823	200	10	39	6	31	0.3870967741935484	0.2326462913533036	0.633303866223915	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, ... (10 total)
637	687	50	4	45	2	35	0.2857142857142857	0.2482159376238932	0.605170336455772	EVT-P-0019922-T01-IM6-558, EVT-P-0027944-T01-IM6-535, EVT-P-0039262-T01-IM6-529, EVT-P-0065470-T01-IM7-545
969	994	25	20	29	17	20	2.55	0.31277927391835814	0.5047620328535949	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (20 total)
297	347	50	5	44	5	32	None	0.31544865864939875	0.5010713149608126	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
876	1076	200	28	21	23	14	2.3	0.3156633225694666	0.5007758765526734	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (28 total)
961	986	25	19	30	16	21	2.2857142857142856	0.3229352452397617	0.49088455337919445	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (19 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
854	1054	200	27	22	22	15	2.0533333333333332	0.3311410503304157	0.4799869780310702	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (27 total)
688	888	200	7	42	4	33	0.36363636363636365	0.3403233440182433	0.46810826004144224	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0030222-T01-IM6-557, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3, ... (7 total)
651	851	200	8	41	5	32	0.46875	0.3858206664977805	0.41361451306963637	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (8 total)
986	1186	200	10	39	9	28	3.5357142857142856	0.4143055366655828	0.38267926287056014	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, ... (10 total)
986	1086	100	9	40	8	29	3.0344827586206895	0.42030696000904744	0.37643341830173344	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, ... (9 total)
991	1091	100	9	40	8	29	3.0344827586206895	0.42030696000904744	0.37643341830173344	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, ... (9 total)
958	983	25	18	31	15	22	2.0454545454545454	0.4941707358480989	0.3061229765300417	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0025018-T01-IM6-539, ... (18 total)
875	1075	200	29	20	23	14	1.6428571428571428	0.5123352016371653	0.2904458034998809	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (29 total)
297	322	25	4	45	4	33	None	0.5599312805603278	0.2518652699665775	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
688	713	25	4	45	4	33	None	0.5599312805603278	0.2518652699665775	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3
1029	1054	25	4	45	4	33	None	0.5599312805603278	0.2518652699665775	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
983	1008	25	5	44	3	34	0.4411764705882353	0.5843795427514207	0.23330499572487204	EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0037526-T01-IM6-525, EVT-P-0037759-T01-IM6-542
663	688	25	6	43	4	33	0.6060606060606061	0.6259415884762788	0.2034661923263583	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (6 total)
247	297	50	6	43	4	33	0.6060606060606061	0.6259415884762788	0.2034661923263583	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
1003	1103	100	8	41	7	30	2.566666666666667	0.6597711605779072	0.18060667193042876	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, ... (8 total)
1038	1138	100	7	42	6	31	2.129032258064516	0.6650872361938774	0.17712138670765892	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-516, ... (7 total)
588	688	100	9	40	6	31	0.5806451612903226	0.6692548912935057	0.1744084460595711	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, ... (9 total)
640	840	200	9	40	6	31	0.5806451612903226	0.6692548912935057	0.1744084460595711	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, ... (9 total)
855	1055	200	28	21	22	15	1.4666666666666666	0.7388569842438368	0.13143961703757803	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, ... (28 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
783	983	200	21	28	15	22	0.6818181818181818	0.7388569842438371	0.1314396170375779	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0025018-T01-IM6-539, ... (21 total)
838	1038	200	24	25	19	18	1.4777777777777779	0.7416433266833333	0.12980490670896044	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (24 total)
782	982	200	19	30	15	22	1.3636363636363635	0.7434280929329625	0.12876103165131966	EVT-P-0015179-T01-IM6-514, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, EVT-P-0028241-T01-IM6-518, ... (19 total)
786	986	200	22	27	16	21	0.7619047619047619	0.7465376611778418	0.12694827818791946	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (22 total)
272	297	25	5	44	4	33	1.3333333333333333	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, EVT-P-0063743-T01-IM7-536
1030	1055	25	5	44	4	33	1.3333333333333333	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
1050	1075	25	5	44	4	33	1.3333333333333333	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0058375-T01-IM6-526
286	336	50	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
638	688	50	8	41	6	31	0.967741935483871	1.0	-0.0	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
651	701	50	7	42	5	32	0.78125	1.0	-0.0	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-535, EVT-P-0036028-T01-IM6-543, EVT-P-0039262-T01-IM6-529, ... (7 total)
1025	1075	50	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0043097-T01-IM6-531, ... (6 total)
1041	1091	50	6	43	5	32	1.71875	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0025191-T02-IM6-522, EVT-P-0029159-T01-IM6-508, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-516, ... (6 total)
236	336	100	7	42	5	32	0.78125	1.0	-0.0	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, ... (7 total)
983	1083	100	11	38	8	29	0.8275862068965517	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0025191-T02-IM6-522, ... (11 total)
97	297	200	7	42	5	32	0.78125	1.0	-0.0	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0028484-T02-IM6-548, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, ... (7 total)
136	336	200	8	41	6	31	0.967741935483871	1.0	-0.0	EVT-P-0016009-T01-IM6-528, EVT-P-0016795-T01-IM6-5, EVT-P-0028484-T02-IM6-548, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, ... (8 total)
794	994	200	23	26	17	20	0.85	1.0	-0.0	EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, EVT-P-0023043-T01-IM6-515, ... (23 total)
803	1003	200	24	25	18	19	0.9473684210526315	1.0	-0.0	EVT-P-0014220-T02-IM6-523, EVT-P-0015179-T01-IM6-514, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-517, EVT-P-0019603-T01-IM6-520, ... (24 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
983	1183	200	12	37	9	28	0.9642857142857143	1.0	-0.0	EVT-P-0005701-T01-IM5-519, EVT-P-0014220-T02-IM6-523, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0025191-T02-IM6-522, ... (12 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003140-T01-IM6-2	P-003140-T01-IM6	P-003140	Breast Cancer	BRCA	BLTP3B-EGFR	BLTP3B	unknown	five_prime	55270272	27	1075	False	1075	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - UHRF1BP1L (NM_015054) Rearrangement :t(7;12)(p11.2;q23.1)(chr7:g.55270272::chr12:g.100460508)| Protein Fusion: mid-exon { EGFR:UHRF1BP1L}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005067 9-T02-IM6-3	P-005067 9-T02-IM6	P-005067 9	Germ Cell Tumor	NSGCT	CADPS-EGFR	CADPS	out-of-frame	three_prime	55241099	18	688	True	688	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase	ReceptorLdomain; Furin-like cysteine rich region; Growth factor receptor domain IV			CADPS (NM_003716) - EGFR (NM_005228) fusion: t(3;7)(p14.2;p11.2) (chr3:g.62537410::chr7:g.55241099)| Protein Fusion: out of frame {CADPS:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003868 3-T01-IM6-4	P-003868 3-T01-IM6	P-003868 3	Melanoma	ARM	CSF2RA-EGFR	CSF2RA	unknown	three_prime	55224220	9	336	True	336	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	ReceptorLdomain	Furin-like cysteine rich region	CSF2RA (NM_001161530) - EGFR (NM_005228) rearrangement: t(7;X)(p11.2;p22.33)(chr7:g.55224220::chrX:g.140166 2) | ProteinFusion: mid-exon {CSF2RA:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001679 5-T01-IM6-5	P-001679 5-T01-IM6	P-001679 5	Breast Cancer	IDC	DDC-EGFR	DDC	out-of-frame	three_prime	55222204	8	297	True	297	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	ReceptorLdomain	Furin-like cysteine rich region	DDC (NM_000790) - EGFR (NM_005228) Rearrangement: c.435+4059:DDC_c.889+359:EGFRinv|Protein Fusion: out of frame {DDC:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001838 9-T02-IM6-6	P-001838 9-T02-IM6	P-001838 9	Colorectal Cancer	COAD	EEA1-EGFR	EEA1	in-frame	three_prime	55241575	18	688	True	688	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase	ReceptorLdomain; Furin-like cysteine rich region; Growth factor receptor domain IV			EEA1 (NM_005366) - EGFR (NM_005228) rearrangement: t(7;12)(p11.2;q22)(chr7:g.55241575::chr12:g.93221089)| Protein Fusion: in frame {EEA1:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006466-T01-IM7-9	P-006466-T01-IM7	P-006466	Bladder Cancer	BLCA	EGFR-CDH7	CDH7	out-of-frame	five_prime	55254206	20	823	True	823	True	exact match	disrupted	0.43873517786561267	True	partner_lacks_domain	PF07714					209			False		ReceptorLdomain; Furin-like cysteine rich region; Growth factor receptor domain IV		Protein tyrosine and serine/threonine kinase	EGFR (NM_005228) - CDH7 (NM_033646) rearrangement: t(7;18)(p11.2;q22.1)(chr7:g.55254206::chr18:g.63486938) | Protein Fusion: out of frame {EGFR:CDH7}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003489 2-T01-IM6-497	P-003489 2-T01-IM6	P-003489 2	Glioma	GB	EGFR-ELAPOR2	ELAPOR2	unknown	three_prime	55268103	24	981	False	981	True	exact match	lost	0.0	False													Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		EGFR (NM_005228) rearrangement: c.2943:EGFR_chr7:g.86587163del | Antisense Fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005788 0-T01-IM6-500	P-005788 0-T01-IM6	P-005788 0	Glioma	GB	EGFR-LANCL2	LANCL2	in-frame	five_prime	55268871	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - LANCL2 (NM_018697) fusion: c.2947-10:EGFR_c.205-8690:LANCL2del | Protein Fusion: in frame {EGFR:LANCL2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005922-T03-IM7-501	P-005922-T03-IM7	P-005922	Glioma	GB	EGFR-LANCL2	LANCL2	in-frame	five_prime	55259704	21	875	True	875	True	exact match	disrupted	0.6442687747035574	True	partner_lacks_domain	PF07714					337			False		ReceptorL domain; Furin-like cysteine rich region; Growth factor receptor domain IV		Protein tyrosine and serine/threonine kinase	EGFR (NM_005228) - LANCL2 (NM_018697) rearrangement: c.2625+137:EGFR_c.1008+3812:LANCL2 del | Protein Fusion: in frame {EGFR:LANCL2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006314 2-T01-IM7 -503	P-006314 2-T01-IM7	P-006314 2	Renal Cell Carcinoma	CCRCC	EGFR-PCDH15	PCDH15	unknown	five_prime	55221820	7	288	False	288	True	exact match	lost	0.0	False													ReceptorLdomain	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Furin-like cysteine rich region	EGFR (NM_005228) - PCDH15 (NM_001142763) rearrangement: t(7;10)(p11.2;q21.1)(chr7:g.55221820::chr10:g.56187463) | Protein Fusion: mid-exon { EGFR:PCDH15 }	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004236-T01-IM6-504	P-004236-T01-IM6	P-004236	Glioma	GB	EGFR-PKD1L1	PKD1L1	in-frame	five_prime	55268824	24	982	True	982	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - PKD1L1 (NM_138295) fusion: c.2947-57: EGFR_c.8527-7825: PKD1L1inv | Protein Fusion: in frame {EGFR:PKD1L1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004114 9-T01-IM6-505	P-004114 9-T01-IM6	P-004114 9	Non-Small Cell Lung Cancer	LUAD	EGFR-RAD51	RAD51	in-frame	five_prime	552687 99	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.55268799::chr15:g.40996189)|Protein Fusion: in frame{EGFR:RAD51}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004114 9-T04-IM6-506	P-004114 9-T04-IM6	P-004114 9	Non-Small Cell Lung Cancer	LUAD	EGFR-RAD51	RAD51	in-frame	five_prime	55268799	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.55268799::chr15:g.40996189)|Protein Fusion: in frame{EGFR:RAD51}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003260 4-T01-IM6-507	P-003260 4-T01-IM6	P-003260 4	Esophagogastric Cancer	ESCA	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268771	24	982	True	982	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion : c.2947-110 :EGFR_c.1120-1204: SEPT14 | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002919-T01-IM6-508	P-002919-T01-IM6	P-002919	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	three_prime	55269850	27	1055	True	1055	True	exact match	lost	0.0	False														Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			SEPT14 (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv | Protein Fusion: in frame {SEPT14: EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003459 4-T01-IM6-509	P-003459 4-T01-IM6	P-003459 4	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268544	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion : c.2947-110 :EGFR_c.1120-1204: SEPT14 | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004352-5-T01-IM6-510	P-004352-5-T01-IM6	P-004352-5	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268859	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22: EGFR_c.1120-807: SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003175 3-T01-IM6-511	P-003175 3-T01-IM6	P-003175 3	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268412	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+306:EGFR_c.1119+2531:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004736-T01-IM6-513	P-004736-T01-IM6	P-004736	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268316	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2946+210:EGFR_c.1120-2817:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001517 9-T01-IM6-514	P-001517 9-T01-IM6	P-001517 9	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268650	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_002529) - SEPT14 (NM_014857) fusion (EGFR exons 1-24 fused with SEPT14 exon 10): c.2947-231:EGFR_c.1119+3910:SEPT14 inv | Protein fusion: in frame (EGFR-SEPT14)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002304 3-T01-IM6-515	P-002304 3-T01-IM6	P-002304 3	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268179	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion (EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+73:EGFR_c.1120-1180:SEPT14 | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003925-T02-IM6-516	P-003925-T02-IM6	P-003925	Esophagogastric Cancer	STAD	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55270807	27	1091	True	1090	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False															Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3271+489:EGFR_c.818-3507:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001625 6-T01-IM6-517	P-001625 6-T01-IM6	P-001625 6	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268390	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion(EGFR exons 1-24 fused to SEPT14 exons 9-10): c.2946+284:EGFR_c.1120-3940:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002824 1-T01-IM6-518	P-002824 1-T01-IM6	P-002824 1	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268525	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2947-356:EGFR_c.1119+4312:SEPT14 inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000570 1-T01-IM5-519	P-000570 1-T01-IM5	P-000570 1	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55269490	26	1054	True	1054	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15: EGFR_c.1119+2833:SEPT14 inv | Protein fusion: in frame (EGFR-SEPT14)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001960 3-T01-IM6-520	P-001960 3-T01-IM6	P-001960 3	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268171	24	982	True	982	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion ((EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+65:EGFR_c.1119+3306:SEPT14 | Protein Fusion : in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003813 8-T01-IM6-521	P-003813 8-T01-IM6	P-003813 8	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268381	24	982	True	982	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+275:EGFR_c.1119+2043:SEPT14 inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002519 1-T02-IM6-522	P-002519 1-T02-IM6	P-002519 1	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55269542	26	1054	True	1054	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3162+67:EGFR_c.1120-3396:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001422 0-T02-IM6-523	P-001422 0-T02-IM6	P-001422 0	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	unknown	five_prime	55268942	25	1003	False	1002	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False															Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-25 fused with SEPT14 exon 10): c.3008:EGFR_c.1120-4102:SEPT14inv | Protein Fusion: mid-exon {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006124 5-T01-IM7-524	P-006124 5-T01-IM7	P-006124 5	Ampullary Cancer	IAMPCA	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268815	24	982	True	982	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement: c.2947-66:EGFR_c.1119+2973:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003752 6-T01-IM6-525	P-003752 6-T01-IM6	P-003752 6	Glioma	GB	EGFR-SEPTIN14	SEPTIN14	unknown	five_prime	55268916	25	994	False	994	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2982:EGFR_c.1119+1151:SEPT14 inv | Protein Fusion: mid-exon { EGFR:SEPT14 }	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005837 5-T01-IM6-526	P-005837 5-T01-IM6	P-005837 5	Endometrial Cancer	UMEC	EGFR-TNRC18	TNRC18	out-of-frame	five_prime	55269897	26	1054	True	1054	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - TNRC18 (NM_001080495) fusion: c.3163-313:EGFR_c.5720-6144:TNRC18inv | Protein Fusion: out of frame {EGFR:TNRC18}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001600 9-T01-IM6-528	P-001600 9-T01-IM6	P-001600 9	Breast Cancer	IDC	EGFR-VOPP1	VOPP1	in-frame	five_prime	55220393	6	249	True	249	True	exact match	lost	0.0	False												ReceptorLdomain	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Furin-like cysteine rich region	EGFR (NM_005228) - VOPP1 (NM_030796): c.747+36:EGFR_c.55-23161VOPP1inv | ProteinFusion: in frame {EGFR:VOPP1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003926-T01-IM6-529	P-003926-T01-IM6	P-003926	Glioma	GB	EGFR-VSTM2A	VSTM2A	out-of-frame	five_prime	55241509	17	687	True	687	True	exact match	lost	0.0	False													ReceptorLdomain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - VSTM2A (NM_182546) rearrangement: c.2062-105:EGFR_c.634+2190:VSTM2A dup | Protein Fusion: out of frame {EGFR:VSTM2A}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004223 6-T01-IM6-530	P-004223 6-T01-IM6	P-004223 6	Glioma	GB	EGFR-VWC2	VWC2	unknown	five_prime	552700225	27	1060	False	1059	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False															Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - VWC2 (NM_198570) fusion: c.3178:EGFR_c.827-13914:VWC2dup | Protein Fusion: mid-exon {EGFR:VWC2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004309 7-T01-IM6-531	P-004309 7-T01-IM6	P-004309 7	Glioma	GB	EGFR-ZNF713	ZNF713	in-frame	five_prime	55269400	25	1038	True	1038	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - ZNF713 (NM_182633) fusion: c.3115-28:EGFR_c.49-962:ZNF713 del | Protein Fusion: in frame {EGFR:ZNF713}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002794-T03-IM6-534	P-002794-T03-IM6	P-002794	Small Cell Lung Cancer	CSCLC	FADD-EGFR	FADD	unknown	five_prime	55241077	17	687	True	687	True	exact match	lost	0.0	False													ReceptorL domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330) | ProteinFusion: mid-exon {EGFR:FADD}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002794-T01-IM6-535	P-002794-T01-IM6	P-002794	Non-Small Cell Lung Cancer	LNET	FADD-EGFR	FADD	unknown	five_prime	55241077	17	687	True	687	True	exact match	lost	0.0	False													ReceptorL domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330) | ProteinFusion: mid-exon {EGFR:FADD}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006374 3-T01-IM7-536	P-006374 3-T01-IM7	P-006374 3	Glioma	GB	GARS1-EGFR	GARS1	out-of-frame	three_prime	55222368	8	297	True	297	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	ReceptorLdomain	Furin-like cysteine rich region	GARS (NM_002047) - EGFR (NM_005228) fusion: c.1032-338:GARS_c.889+523:EGFR del | Protein Fusion: out of frame {GARS:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002103 8-T01-IM6-538	P-002103 8-T01-IM6	P-002103 8	Non-Small Cell Lung Cancer	LUAD	KIF5B-EGFR	KIF5B	in-frame	three_prime	55241486	18	688	True	688	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase	ReceptorLdomain; Furin-like cysteine rich region; Growth factor receptor domain IV		KIF5B (NM_004521) - EGFR (NM_005228) rearrangement: t(7;10)(p11.2;q11.22)(chr7:g.55241486::chr10:g.32312089)|Protein Fusion: in frame {KIF5B:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002501 8-T01-IM6-539	P-002501 8-T01-IM6	P-002501 8	Non-Small Cell Lung Cancer	LUSC	LANCL2-EGFR	LANCL2	out-of-frame	three_prime	55268537	25	983	True	983	True	exact match	lost	0.0	False															Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		LANCL2 (NM_018697) - EGFR (NM_005228) rearrangement: c.1259-255:LANCL2_c.2947-344:EGFRdup | Protein Fusion: out of frame {LANCL2:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003775 9-T03-IM6-540	P-003775 9-T03-IM6	P-003775 9	Breast Cancer	BRCA	NIPSNAP2-EGFR	NIPSNAP2	out-of-frame	three_prime	552684 44	25	983	True	983	True	exact match	lost	0.0	False													Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		GBAS (NM_001483) - EGFR (NM_005228) rearrangement: c.279-1159:GBAS_c.2946+338:EGFRdup | Protein Fusion: out of frame {GBAS:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005922-T01-IM7-541	P-005922-T01-IM7	P-005922	Glioma	GB	NIPSNAP2-EGFR	NIPSNAP2	unknown	three_prime	55233046	15	599	False	599	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase	ReceptorLdomain; Furin-like cysteine rich region	Growth factor receptor domain IV	GBAS (NM_001483) - EGFR (NM_005228) fusion: c.92+717: GBAS_c.1796: EGFRdup | Protein Fusion: mid-exon {GBAS:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003775 9-T01-IM6-542	P-003775 9-T01-IM6	P-003775 9	Breast Cancer	IDC	NIPSNAP2-EGFR	NIPSNAP2	out-of-frame	three_prime	552684 44	25	983	True	983	True	exact match	lost	0.0	False													Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		GBAS (NM_001483) - EGFR(NM_005228) Rearrangement: c.2946+338:EGFR_c.279-1159:GBASdup | Protein Fusion: out of frame {GBAS:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003602 8-T01-IM6-543	P-003602 8-T01-IM6	P-003602 8	Colorectal Cancer	COAD	NUMA1-EGFR	NUMA1	in-frame	three_prime	55241128	18	688	True	688	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase	ReceptorLdomain; Furin-like cysteine rich region; Growth factor receptor domain IV			NUMA1 (NM_006185) - EGFR (NM_005228) rearrangement: t(7;11)(p11.2;q13.4)(chr7:g.55241128::chr11:g.71728415) | Protein Fusion: in frame {NUMA1:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006547 0-T01-IM7-545	P-006547 0-T01-IM7	P-006547 0	Glioma	GB	PDE1C-EGFR	PDE1C	unknown	three_prime	55240707	17	651	False	651	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase	ReceptorLdomain; Furin-like cysteine rich region; Growth factor receptor domain IV		PDE1C (NM_001191058) - EGFR (NM_005228) fusion: c.86-15224:PDE1C_c.1951:EGFRinv | Protein Fusion: mid-exon {PDE1C:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005903 2-T01-IM7-546	P-005903 2-T01-IM7	P-005903 2	Non-Small Cell Lung Cancer	LUAD	PLOD3-EGFR	PLOD3	out-of-frame	three_prime	55179084	2	30	True	30	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			PLOD3 (NM_001084) - EGFR (NM_005228) fusion: c.2062-9:PLOD3_c.89-30895:EGFRinv|Protein Fusion: out of frame{PLOD3:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002848 4-T02-IM6-548	P-002848 4-T02-IM6	P-002848 4	Bone Cancer	OS	SCAF4-EGFR	SCAF4	out-of-frame	three_prime	552141 45	4	142	True	142	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Furin-like cysteine rich region; Growth factor receptor domain IV		ReceptorLdomain	SCAF4 (NM_020706) - EGFR (NM_005228) fusion: t(7;21)(p11.2;q22.11)(chr7:g.55214145::chr21:g.33094864) | Protein Fusion: out of frame {SCAF4:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00309100-T01-IM6-551	P-00309100-T01-IM6	P-00309100	Pancreatic Cancer	PAAC	SEL1L-EGFR	SEL1L	in-frame	three_prime	55222485	8	297	True	297	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	ReceptorLdomain	Furin-like cysteine rich region	SEL1L (NM_005065) - EGFR (NM_005228) rearrangement: t(7;14)(p11.2;q31.1)(chr7:g.55222485::chr14:g.81986576) | Protein Fusion: in frame {SEL1L:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005297 6-T02-IM7-552	P-005297 6-T02-IM7	P-005297 6	Glioma	GB	TNS3-EGFR	TNS3	in-frame	three_prime	55222307	8	297	True	297	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	ReceptorLdomain	Furin-like cysteine rich region	TNS3 (NM_022748) - EGFR (NM_005228) fusion: c.1025-2026:TNS3_c.889+462:EGFRinv|Protein Fusion: in frame {TNS3:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000687 3-T01-IM5-554	P-000687 3	P-000687 3	Non-Small Cell Lung Cancer	LUSC	TUT7-EGFR	TUT7	unknown	five_prime	55238031	15	627	True	626	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False															ReceptorLdomain; Furin-like cysteine rich region	Protein tyrosine and serine/threonine kinase	Growth factor receptor domain IV	EGFR (NM_005228) - ZCCHC6 (NM_024617) rearrangement : : t(7;9)(chr7p11.2;chr9q21.33)(chr7:g.55238031::chr9:g.88930920) | Protein fusion: mid-exon (EGFR-ZCCHC6)	NA

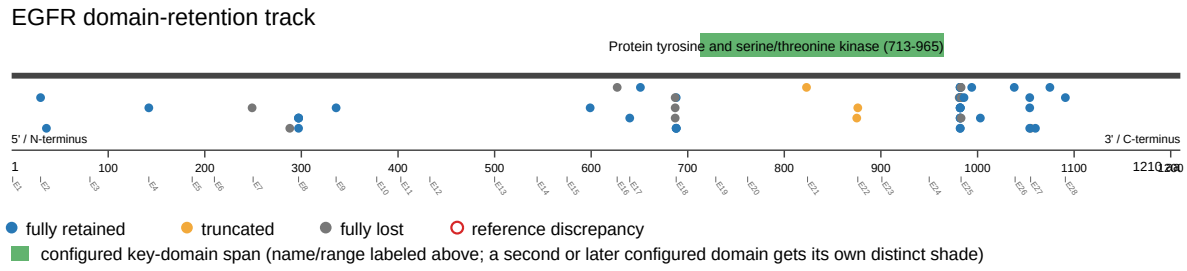
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001541 9-T01-IM6-555	P-001541 9-T01-IM6	P-001541 9	Nerve Sheath Tumor	MPNST	VPS41-EGFR	VPS41	unknown	five_prime	55268890	25	986	False	985	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False														Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - VPS41 (NM_014396) Rearrangement : c.2956:EGFR_c.247-4922:VPS41inv | Protein fusion: mid-exon (EGFR-VPS41)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003022-T01-IM6-557	P-003022-T01-IM6	P-003022	Glioma	GB	VWC2-EGFR	VWC2	out-of-frame	three_prime	55259621	22	876	True	876	True	exact match	disrupted	0.3557312252964427	True	partner_lacks_domain	PF07714					276			False			ReceptorL domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase	VWC2 (NM_198570) - EGFR (NM_005228) rearrangement: c.827-37789:VWC2_c.2625+54:EGFRdel|Protein Fusion: out of frame{VWC2:EGFR}	NA
EVT-P-001992-T01-IM6-558	P-001992-T01-IM6	P-001992	Soft Tissue Sarcoma	DDLS	YIF1B-EGFR	YIF1B	unknown	three_prime	55238939	17	640	True	640	True	exact match	retained	1.0	False												Protein tyrosine and serine/threonine kinase	ReceptorL domain; Furin-like cysteine rich region; Growth factor receptor domain IV		EGFR (NM_005228) rearrangement: t(7;19)(p22.1;q13.2)(chr7:g.55238939::chr19:g.38806683) | Antisense Fusion	NA

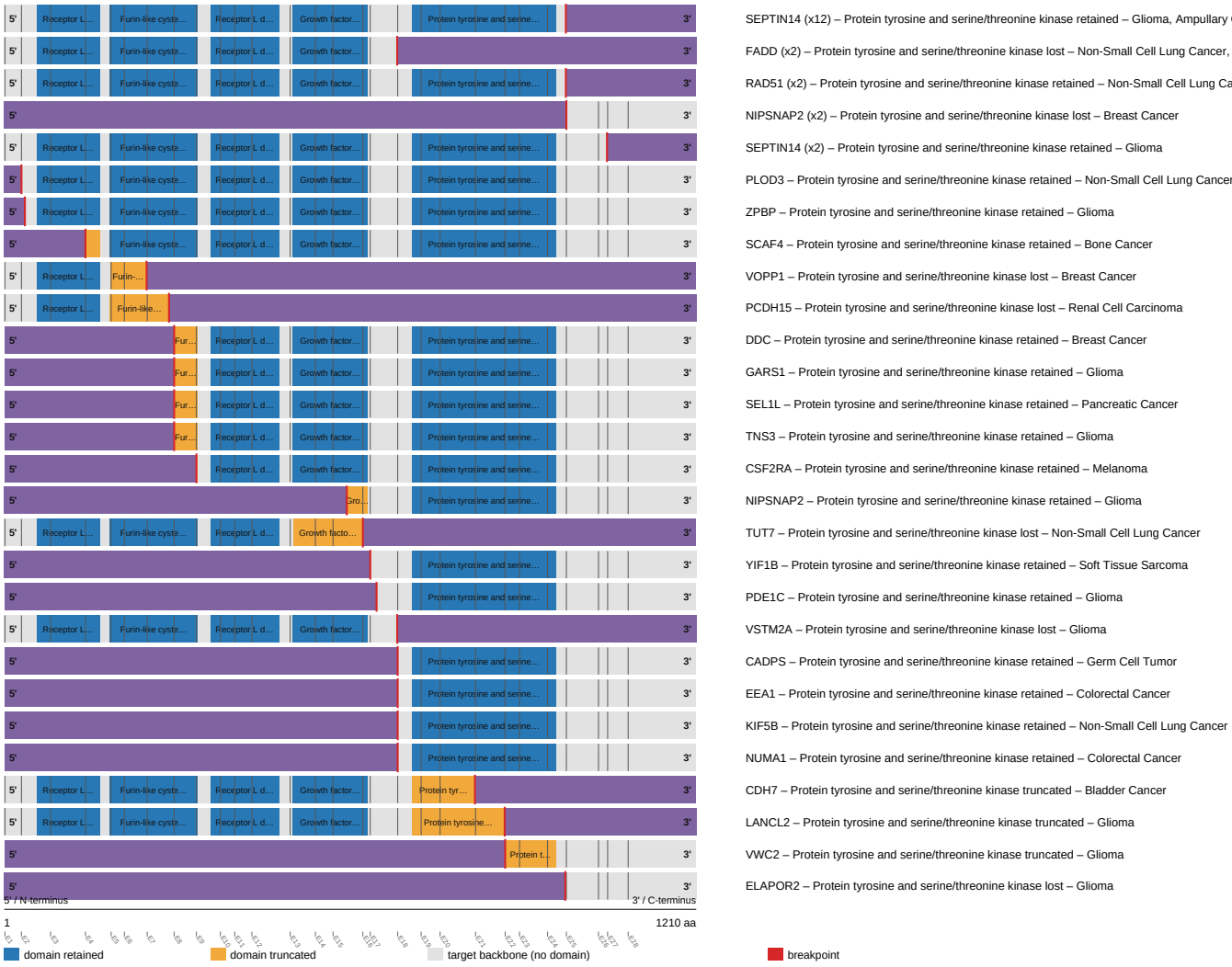
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003022-T01-IM6-559	P-003022-T01-IM6	P-003022	Glioma	GB	ZBP-EGFR	ZBP	unknown	three_prime	55209996	2	36	False	36	True	exact match	retained	1.0	False													Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			ZBP (NM_007009) - EGFR (NM_005228) rearrangement: c.487+5719:ZBP_c.106:EGFRinv | Protein Fusion: mid-exon {ZBP:EGFR}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 40 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic

